Engineering Economy

Chapter 3:

Cost Estimation Techniques

Chapter 3
Cost Estimation Techniques

The objective of Chapter 3
is to present
various methods for estimating important factors
in an engineering economy study.

Chapter 3
Cost Estimation Techniques

Engineering Economic Analysis Procedure

- Problem definition
- Development of alternatives
- Development of prospective outcomes
- Selection of a decision criterion
- Analysis and comparison of alternatives.
- Selection of the preferred alternative.
- Performance monitoring and postevaluation of results.

Chapter 3
3.1. Introducton

Estimating the future cash flows for feasible alternatives is a critical step in engineering economy studies.

Estimating costs, revenues, useful lives, residual values, and other pertinent data can be the most difficult, expensive, and time-consuming part of the study.

Chapter 3
3.1. Introducton

Results of cost estimating are used for a variety of purposes.

- Setting selling prices for quoting, bidding, or evaluating contracts.
- Determining if a proposed product can be made and distributed at a profit.
- Evaluating how much capital can be justified for changes and improvements.
- Setting benchmarks for productivity improvement programs.

Chapter 3
3.1. Introducton

The two fundamental approaches are "top-down" and "bottom-up."

- **Top-down** uses historical data from similar projects. It is best used when alternatives are still being developed and refined.

- **Bottom-up** is more detailed and works best when the detail concerning the desired output (product or service) has been defined and clarified.

Chapter 3
Example 3-1

Estimating the Cost of a College Degree

A simple example of cost estimating is to forecast the expense of getting a Bachelor of Science (B.S.) from the university you are attending. In our solution, we outline the two basic approaches just discussed for estimating these costs.

Chapter 3
Solution 3-1

Solution

A top-down approach would take the published cost of a four-year degree at the same (or a similar) university and adjust it for inflation and extraordinary items that an incoming student might encounter, such as fraternity/sorority membership, scholarships, and tutoring. For example, suppose that the published cost of attending your university is $15,750 for the current year. This figure is anticipated to increase at the rate of 6% per year and includes full-time tuition and fees, university housing, and a weekly meal plan. Not included are the costs of books, supplies, and other personal expenses. For our initial estimate, these "other" expenses are assumed to remain constant at $5,000 per year.

The total estimated cost for four years can now be computed. We simply need to adjust the published cost for inflation each year and add in the cost of "other" expenses.

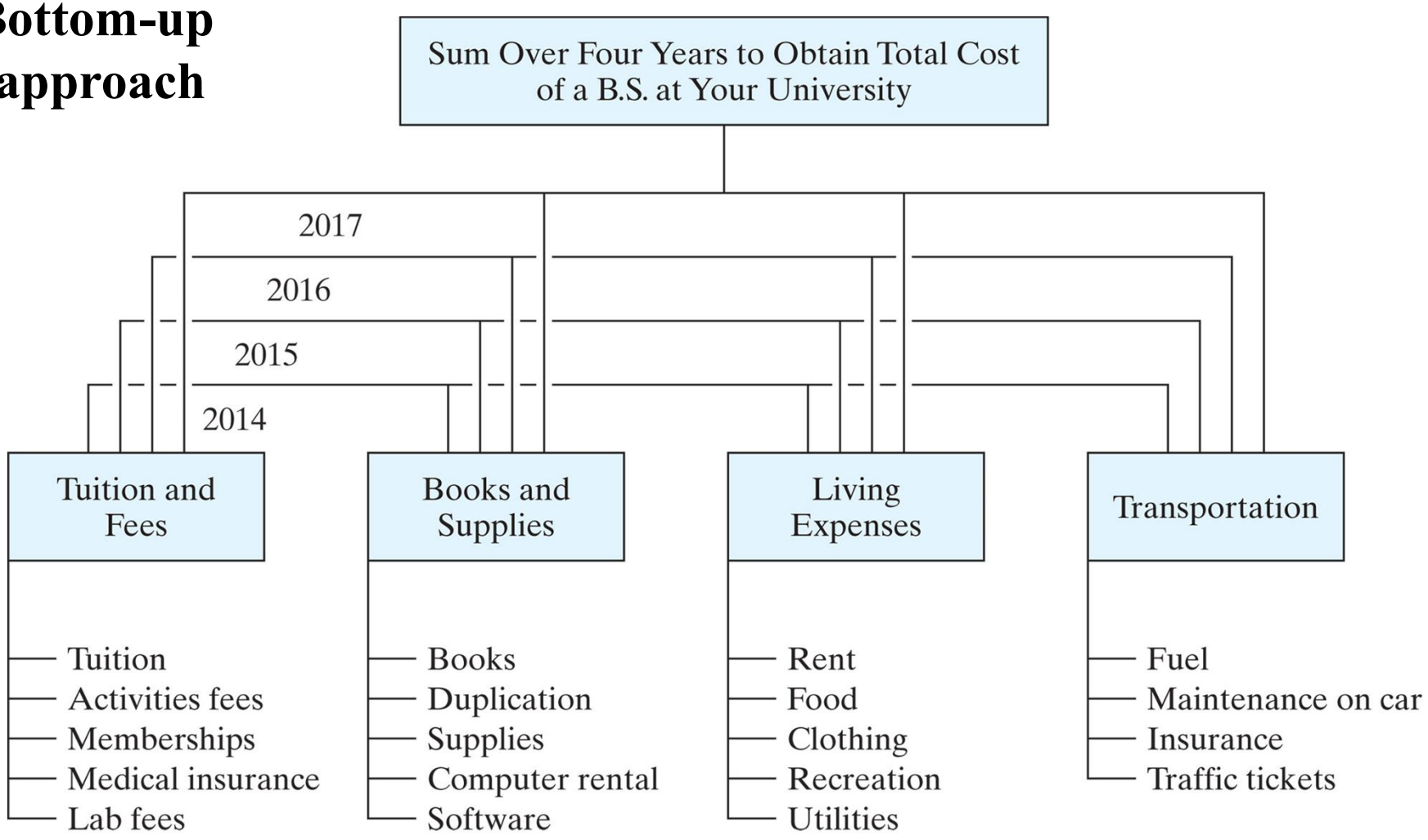
Chapter 3
Solution 3-1

Estimating the Cost of a College Degree

Year	Tuition, Fees, Room and Board	"Other" Expenses	Total Estimated Cost for Year
1	$15,750 × 1.06 = $16,695	$5,000	$21,695
2	16,695 × 1.06 = 17,697	5,000	22,697
3	17,697 × 1.06 = 18,759	5,000	23,759
4	18,759 × 1.06 = 19,885	5,000	24,885
		Grand Total	$93,036

Chapter 3
Solution 3-1

Bottom-up approach



Copyright ©2015 Pearson Education, All Rights Reserved

Chapter 3
Solution 3-1

In contrast with the top-down approach, a bottom-up approach to the same cost estimate would be to first break down anticipated expenses into the typical categories shown in Figure 3-1 for each of the four years at the university. Tuition and fees can be estimated fairly accurately in each year, as can books and supplies. For example, suppose that the average cost of a college textbook is $100. You can estimate your annual textbook cost by simply multiplying the average cost per book by the number of courses you plan to take. Assume that you plan on taking five courses each semester during the first year. Your estimated textbook costs would be

$$\left(\frac{5 \text{ courses}}{\text{semester}}\right)(2 \text{ semesters})\left(\frac{1 \text{ book}}{\text{course}}\right)\left(\frac{\$100}{\text{book}}\right) = \$1{,}000.$$

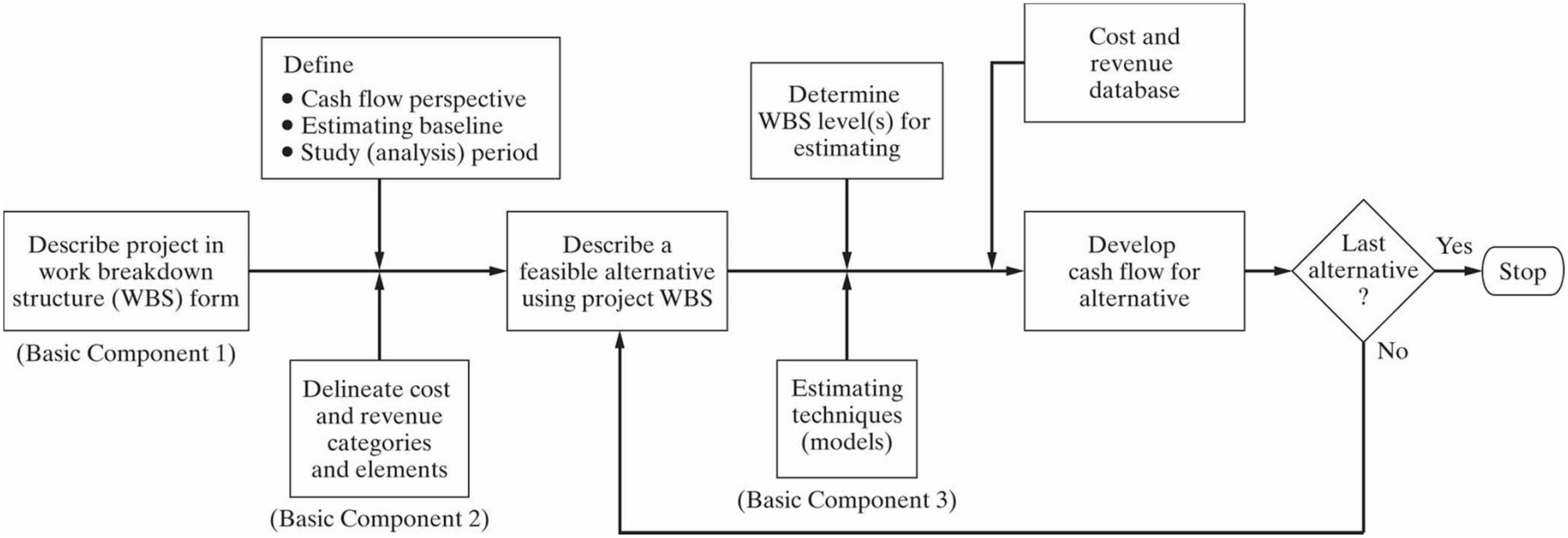
The other two categories, living expenses and transportation, are probably more dependent on your lifestyle. For example, whether you own and operate an automobile and live in a "high-end" apartment off-campus can dramatically affect the estimated expenses during your college years.

Chapter 3
3.2. An Integrated Approach

The integrated cost estimation approach has three major components.

- Work breakdown structure (WBS)
- Cost and revenue structure (classification)
- Estimating techniques (models)

Chapter 3
3.2. An Integrated Approach

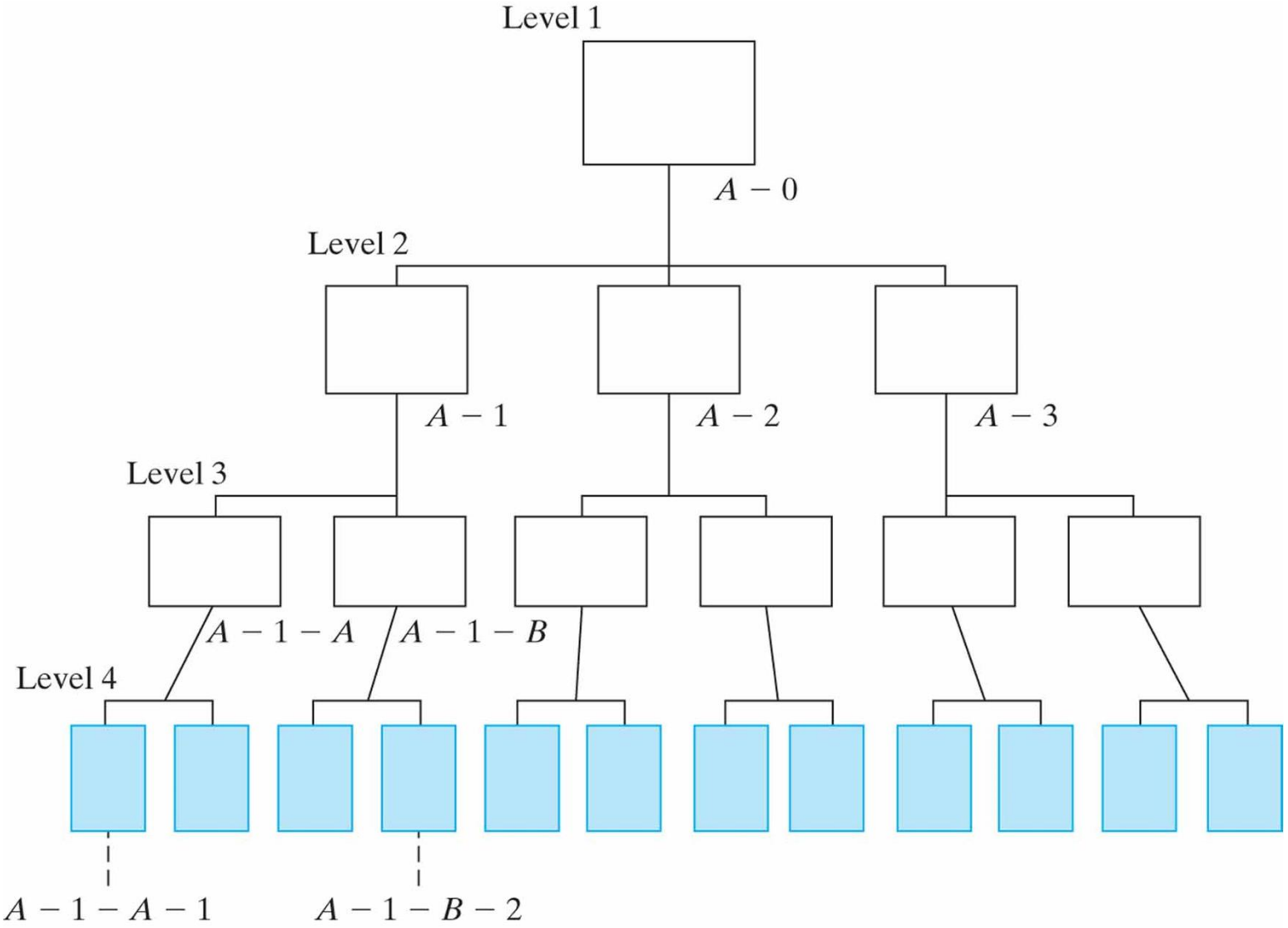


Chapter 3
3.2. An Integrated Approach

3.2.1 Work Breakdown Structure (WBS)

- A basic tool in project management
- A framework for defining all project work elements and their relationships, collecting and organizing information, developing relevant cost and revenue data, and management activities.
- Each level of a WBS divides the work elements into increasing detail.

Chapter 3
3.2. An Integrated Approach



Chapter 3
3.2. An Integrated Approach

A WBS has other characteristics.

- Both functional and physical work elements are included.
- The content and resource requirements for a work element are the sum of the activities and resources of related subelements below it.
- A project WBS usually includes recurring and nonrecurring work elements.

Chapter 3
Example 3-2

A WBS for a Construction Project

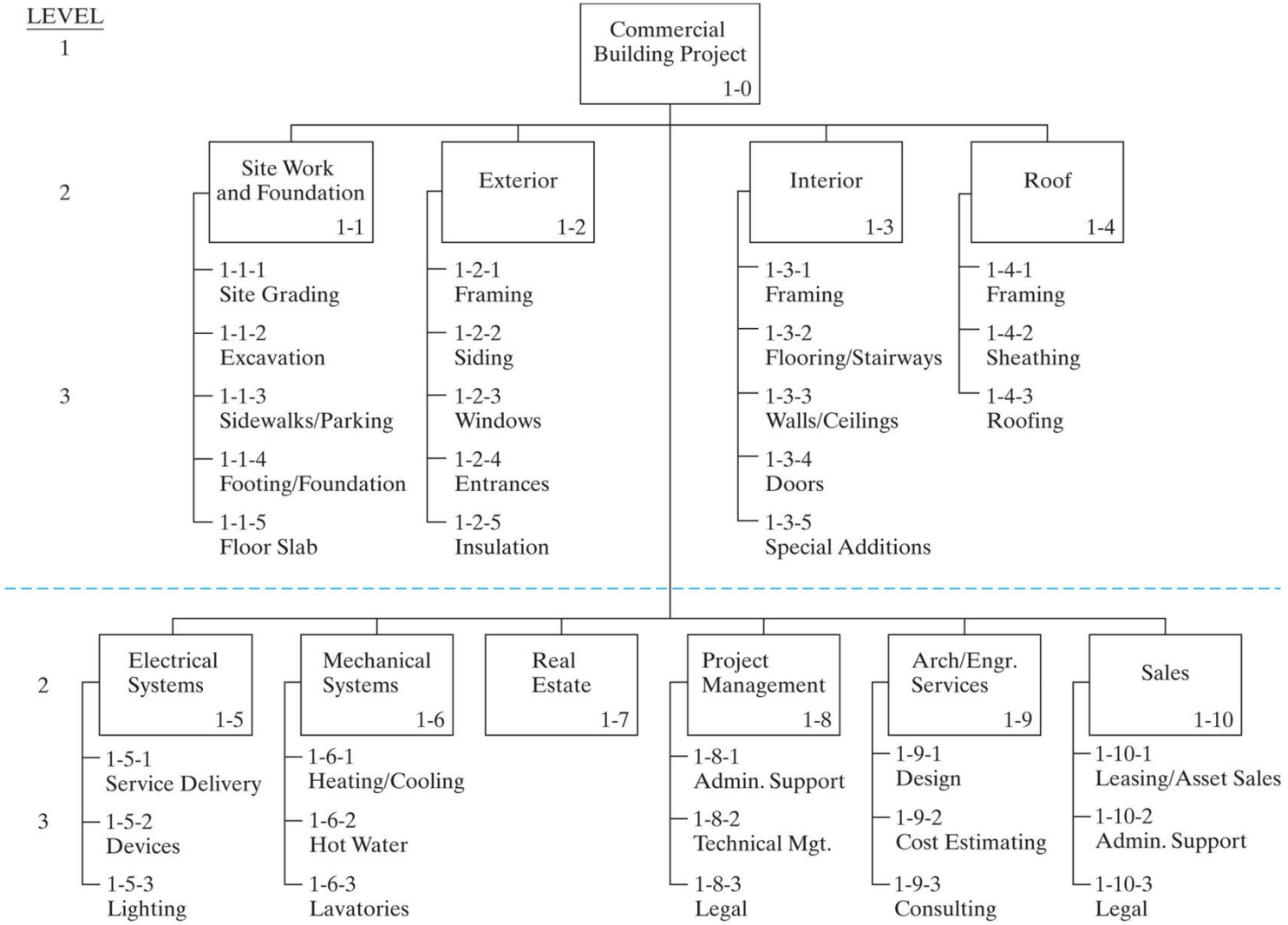
You have been appointed by your company to manage a project involving construction of a small commercial building with two floors of 15,000 gross square feet each. The ground floor is planned for small retail shops, and the second floor is planned for offices. Develop the first three levels of a representative WBS adequate for all project efforts from the time the decision was made to proceed with the design and construction of the building until initial occupancy is completed.

Chapter 3
Solution 3-2

Solution

There would be variations in the WBSs developed by different individuals for a commercial building. A representative three-level WBS is shown in Figure 3-4. Level 1 is the total project. At Level 2, the project is divided into seven major physical work elements and three major functional work elements. Then each of these major elements is divided into subelements as required (Level 3). The numbering scheme used in this example is all numeric.

Chapter 3
Solution 3-2



Copyright ©2015 Pearson Education, All Rights Reserved

Chapter 3
3.2. An Integrated Approach

3.2.2 Cost and Revenue Structure

- Used to identify and categorize the costs and revenues that need to be included in the analysis.
- The life-cycle concept and WBS are important aids in developing the cost and revenue structure for a project.
- Perhaps the most serious source of errors in developing cash flows is overlooking important categories of costs and revenues.

Chapter 3
3.2. An Integrated Approach

The following is a brief listing of some categories of costs and revenues that are typically needed in an engineering economy study:

- Capital investment
- Labor costs
- Material costs
- Maintanence costs
- Property taxes and insurance
- Overhead costs
- Disposal costs
- Revenue based on sales, ect.
- Quality costs
- Market values

Chapter 3
3.2. An Integrated Approach

3.2.3 Estimating Techniques (Models)

Used to develop individual cash-flow estimates and the overall net cash flow for each alternatives

Cost and revenue estimates can be classified according to detail, accuracy, and their intended use as follows:

- Order-of-magnitude estimates (±30%)
- Semidetailed, or budget, estimates (±15%)
- Definitive (detailed) estimates (±5%)

Chapter 3
3.2. An Integrated Approach

Estimating Techniques

REMEMBER! The purpose of estimating is to develop cash-flow projections—*not* to produce exact data about the future, which is virtually impossible.

Chapter 3
3.2. An Integrated Approach

The level of detail and accuracy of estimates depends on

- time and effort available as justified by the importance of the study,
- difficulty of estimating the items in question,
- methods or techniques employed,
- qualifications of the estimator(s), and
- sensitivity of study results to particular factor estimates.

Chapter 3
3.2. An Integrated Approach

A variety of sources exist for cost and revenue estimation.

- Accounting records: good for historical data, but limited for engineering economic analysis.
- Other sources inside the firm: e.g., sales, engineering, production, purchasing.
- Sources outside the firm: government data, industry surveys, trade journals, and personal contacts.
- Research and development: e.g., pilot plant, test marketing program, surveys.

Chapter 3
3.3. Selected Estimating Techniques (Models)

These models can be used in many types of estimates.

- Indexes
- Unit technique
- Factor technique

Chapter 3
3.3. Selected Estimating Techniques (Models)

3.3.1 Indexes, *I*, provide a means for developing present and future cost and price estimates from historical data.

$$C_n = C_k \left(\frac{\bar{I}_n}{\bar{I}_k} \right)$$

k = reference year for which cost or price is known.
n = year for which cost or price is to be estimated ($n>k$).
C_n = estimated cost or price of item in year n.
C_k = cost or price of item in reference year k.

Indexes can be created for a single item or for multiple items (eqs. 3-1, 3-2).

Chapter 3
Example 3-3

Indexing the Cost of a New Boiler

A certain index for the cost of purchasing and installing utility boilers is keyed to 1988, where its baseline value was arbitrarily set at 100. Company XYZ installed a 50,000-lb/hour boiler for $525,000 in 2000 when the index had a value of 468. This same company must install another boiler of the same size in 2014. The index in 2014 is 542. What is the approximate cost of the new boiler?

Solution

In this example, n is 2014 and k is 2000. From Equation (3-1), an approximate cost of the boiler in 2014 is

$$C_{2014} = \$525{,}000(542/468) = \$608{,}013.$$

Chapter 3
Example 3-4

In 2002 Acme Chemical purchased a large pump for $112,000. Acme keys their cost estimating for these pumps to the industrial pump index, with a baseline of 100 established in 1992. The index in 2002 was 212. Acme is now (2010) considering construction of a new addition and must estimate the cost of the same type and size of pump. If the industrial pump index is currently 286, what is the estimated cost of the new pump?

Chapter 3
Solution 3-4

Cost of the new pump (using eq. 3-1)

$$C_n = C_k \left(\frac{\bar{I}_n}{\bar{I}_k} \right)$$

$$\$112,000 \left(\frac{286}{212} \right) = \$151,094$$

Chapter 3
3.3. Selected Estimating Techniques (Models)

3.3.2. The unit technique is one that is widely known and understood.

A "per unit factor" is used, along with the appropriate number of units, to find the total estimate of cost.

Such as;

- Revenue per mile
- Capital cost of plant per kilowatt of capacity
- Capital cost per installed telephone
- Operating cost per mile

This techniques is useful in preliminary estimates, but using average costs can be very misleading.

Chapter 3
Example 3-5

An often used example is the cost of a particular house.

Using a per unit factor of, say, $120 per square foot, and applying that to a house with 3,000 square feet,

results in an estimated cost of $120 x 3,000 = $360,000.

Chapter 3
3.3. Selected Estimating Techniques (Models)

3.3.3. The factor technique is an extension of the unit technique where the products of several quantities are summed and then added to components estimated directly.

$$C = \sum_{d} C_d + \sum_{m} f_m U_m$$

C = cost being estimated
C_d = cost of the selected component d estimated directly
f_m = cost per unit of component m
U_m = number of units of component m

Chapter 3
Example 3-6

We need to estimate the cost of a house of 2000 square feet; two porches, and a garage.

Using a unit factor of $85 per square foot, $10,000 per porch, and $8,000 per garage

Total cost estimation is:

2000 square feet house	-	$85 per square foot
2 porches	-	$10,000 per porch
1 garage	-	$8,000 per garage

C = (2000 x $85) + (2 x $10,000) + (1 x $8,000)
C = $198,000

Chapter 3
3.4. Parametric Cost Estimating

Parametric cost estimating

It uses historical cost data and statistical techniques to predict future costs.

Parametric models are used in the early design stages to get an idea of how much the product (or project) will cost, on the basis of a few physical attributes (such as weight, volume, and power).

Chapter 3
3.4. Parametric Cost Estimating

3.4.1. The power-sizing technique (or *exponential model*) is frequently used for developing capital investment estimates for industrial plants and equipment.

$$C_A = C_B \left(\frac{S_A}{S_B} \right)^X$$

$C_A =$ cost for plant A
$C_B =$ cost for plant B

(both in $ as of the point in time for which the estimate is desired)

$S_A =$ size of plant A
$S_B =$ size of plant B

(both in the same physical units)

$X = cost - capacity factor$ to reflect economies of scale.

Chapter 3
Example 3-7

Acme Logistics provides "less than truck load" (LTL) services throughout the U.S. They have several hubs where they use cross-docking to move goods from one trailer to another.

Acme built its last hub 10 years ago, and it had *36 dock doors*. *The cost index* at that time was *140*, and the *total cost* was *$6 million*.

Acme plans a new hub that will have *48 dock doors*. *The cost index* now is *195*, and Acme will use a *capacity factor of 0.82*.

What is the estimated cost of the new hub?

Chapter 3
Solution 3-7

Cost of the new hub (using eqs. 3-1 and 3-4)

$$C_n = C_k \left(\frac{\bar{I}_n}{\bar{I}_k} \right)$$

$$C_B = \$6,000,000 \left(\frac{195}{140} \right) = \$8,357,143$$

$$C_A = C_B \left(\frac{S_A}{S_B} \right)^X$$

$$C_A = \$8,357,143 \left(\frac{48}{36} \right)^{0.82} = \$10,580,534$$

Chapter 3
3.4. Parametric Cost Estimating

3.4.2 A learning curve reflects increased efficiency and performance with repetitive production of a good or service. The concept is that some input resources decrease, on a per-output-unit basis, as the number of units produced increases.

Most learning curves assume a constant percentage reduction occurs as the number of units produced is *doubled*.

$$Z_u = K(u^n)$$

u = the output unit number,

Z_u = number of input units needed to produce output unit u,

K = number of input units needed to produce the first output unit,

s = the learning curve slop parameter as a decimal,

$n = \frac{log\ s}{log\ 2}$ = the learning curve exponent.

Chapter 3
Example 3-8

Assume the first unit of production required 3 hours time for assembly.
The learning rate is 75%.

Find (a) the time to assemble the 8th unit,
(b) the time needed to assemble the first 6 units.

Chapter 3
Solution 3-8

(a) the time to assemble the 8th unit

$$Z_u = K(u^n)$$

$$Z_8 = 3\ (8^{(log\ 0.75/log\ 2)})$$

$$Z_8 = 1.27 \text{ hours}$$

(b) the time needed to assemble the first 6 units

$$T_x = K\sum_{u=1}^{x} u^n$$

$$T_6 = 3\sum_{u=1}^{6} u^{(log\ 0.75/log\ 2)}$$

$$T_6 = 3(3.9346) = 11.8 \text{ hours}$$

Chapter 3
Example 3-9

Learning Curve for a Formula Car Design Team

The Mechanical Engineering department has a student team that is designing a formula car for national competition. The time required for the team to assemble the first car is 100 hours. Their improvement (or learning rate) is 0.8, which means that as output is doubled, their time to assemble a car is reduced by 20%. Use this information to determine

(a) the time it will take the team to assemble the 10th car.

(b) the *total time* required to assemble the first 10 cars.

(c) the estimated *cumulative average* assembly time for the first 10 cars.

Solve by hand and by spreadsheet.

Chapter 3
Solution 3-9

Learning Curve for a Formula Car Design Team

Solution by Hand

(a) From Equation (3-5), and assuming a proportional decrease in assembly time for output units between doubled quantities, we have

$$Z_{10} = 100(10)^{\log 0.8/\log 2}$$

$$= 100(10)^{-0.322}$$

$$= \frac{100}{2.099} = 47.6 \text{ hours.}$$

(b) The total time to produce x units, T_x, is given by

$$T_x = \sum_{u=1}^{x} Z_u = \sum_{u=1}^{x} K(u^n) = K \sum_{u=1}^{x} u^n. \qquad (3\text{-}6)$$

Using Equation (3-6), we see that

$$T_{10} = 100 \sum_{u=1}^{10} u^{-0.322} = 100[1^{-0.322} + 2^{-0.322} + \cdots + 10^{-0.322}] = 631 \text{ hours.}$$

Chapter 3
Solution 3-9

(c) The cumulative average time for x units, C_x, is given by

$$C_x = T_x/x. \tag{3-7}$$

Using Equation (3-7), we get

$$C_{10} = T_{10}/10 = 631/10 = 63.1 \text{ hours.}$$

Chapter 3
3.4. Parametric Cost Estimating

3.4.3. Developing a Cost Estimating Relationship

A cost estimating relationship (CER) describes the cost of a project as a function of design variables.

There are four basic steps in developing a CER

- Problem definition
- Data collection and normalization
- CER equation development
- Model validation and documentation

Chapter 3
Cost Estimation Techniques

$$C_n = C_k \left(\frac{\bar{I}_n}{\bar{I}_k} \right)$$

Indexes – cost estimation technique

$$C = \sum_d C_d + \sum_m f_m U_m$$

Factor technique – cost estimation technique

$$C_A = C_B \left(\frac{S_A}{S_B} \right)^X$$

Power-sizing technique – cost estimation technique

$$Z_u = K(u^n)$$

Learning curve calculations

$$Tx = K \sum_{u=1}^{x} u^n$$

Total time requirement for production